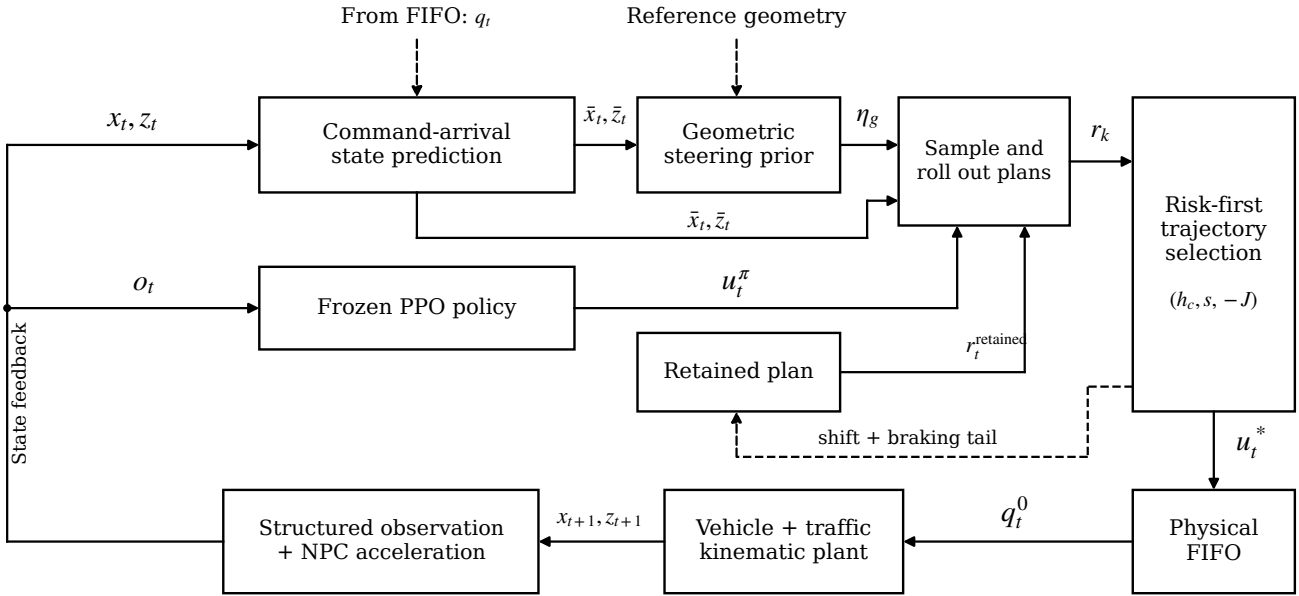
From FIFO: q_t
Reference geometry
x_t, z_t
Command-arrival state prediction
$\bar{x}_t, \bar{z}_t$
Geometric steering prior
η_g
Sample and roll out plans
r_k
Risk-first trajectory selection
$(h_c, s, -J)$
$\bar{x}_t, \bar{z}_t$
o_t
Frozen PPO policy
u_t^π
Retained plan
r_t^{retained}
shift + braking tail
u_t^*
State feedback
Structured observation + NPC acceleration
x_{t+1}, z_{t+1}
Vehicle + traffic kinematic plant
q_t^0
Physical FIFO